

Book of Abstracts

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Table of Contents

<i>Some Innovative Type Linguistic Linguistic Interval-Valued Intuitionistic Fuzzy Choquet Integral Operators Based On Aczel-Alsina T-Norm And T-Conorm With Application To Decision Making</i>	<i>2</i>
<i>Numerical Study Of Partial Differential Equations Using Haar Wavelet Technique.....</i>	<i>3</i>
<i>Solving Higher Order And Multi-Dimensional Partial Differential Equations Via Scale-3 Haar Wavelets</i>	<i>4</i>
<i>A Real-Time Large-Scale IoT Traffic Anomalies Detection System Using Auxiliary Classifier Generative Adversarial Network.....</i>	<i>5</i>
<i>Incentivizing Cooperation and Community Formation in Social Internet of IoT Networks: A study on Virtual Currencies and Social Behaviors</i>	<i>6</i>

Some Innovative Type Linguistic Interval-Valued Intuitionistic Fuzzy Choquet Integral Operators Based On Aczel- Alsina T-Norm And T-Conorm With Application To Decision Making

Afra Siab

Department of Mathematics Kohat University of Science and Technology Pakistan

Abstract:

This article integrates two methodologies Aczel-Alsina (AA) t-norm and t-conorm, along with the Choquet integral (CI) to develop several innovative linguistic interval-valued intuitionistic fuzzy aggregation operators. Since Aczel-Alsina norms generalize several other norms, the study first establishes a set of operational rules for linguistic interval-valued intuitionistic fuzzy numbers. Based on these rules, various aggregation operators are proposed, including the Linguistic Interval-Valued Intuitionistic Fuzzy Choquet Integral Aczel-Alsina Averaging (LIVIFC-IAAA) operator, LIVIFCI-AA Geometric (LIVIFC-IAAG) operator. The proposed operators are not only flexible but also deal the correlation among criteria. Properties of the proposed operators are discussed in details. Further, a decision making approach is developed based on the proposed operators. Moreover, a real-world example is presented for the application and effectiveness of the developed approach. Finally the proposed approach is compared with existing methods

Numerical Study Of Partial Differential Equations Using Haar Wavelet Technique

Muhammad Fiaz, Dr. Abdul Ghafoor, Dr. Emel Khan

Department of Mathematics Kohat University of Science and Technology Pakistan

Abstract:

This work introduces an effective numerical scheme for solving three-dimensional advection-diffusion equations using the Haar wavelet method. In this approach, Haar wavelets are employed to discretized the spatial domain, while conventional finite difference techniques handle the temporal evolution. The highest-order spatial derivatives are approximated through a truncated Haar wavelet series, and integration is used to derive lower-order derivatives and the solution. The differential equations are converted into a system of linear algebraic equations using a collocation approach, which allows for iterative determination of the wavelet coefficients. Theoretical analysis demonstrates that the proposed method is both convergent and stable. Its performance is validated through numerical experiments on two advection problems and one diffusion problem. The errors are evaluated using various norms, including L_∞ , L_2 , and L_{RMS} , as well as relative error metrics such as RE_∞ and RE_2 . The results indicate that the method accurately captures the solution behavior and achieves a high level of agreement with analytical solutions. This highlights the efficiency and reliability of the Haar wavelet method for solving complex three-dimensional advection-diffusion equations.

Solving Higher Order And Multi-Dimensional Partial Differential Equations Via Scale-3 Haar Wavelets

Muhammad Bilal, Dr. Abdul Ghafoor, Dr. Emel Khan

Department of Mathematics, Kohat University of Science and Technology Pakistan

Abstract:

This research work presents an efficient and robust numerical estimation strategy combining scale-3 Haar wavelet with finite difference formulation to effectively solve two-dimensional time dependent partial differential equations. The temporal derivative is approximated by finite difference while the spatial derivatives along with solution is estimated by scale-3 Haar wavelet. Few test problems are presented to confirm the accuracy and effectiveness of the proposed methodology. The outcomes are matched with exact solutions and those with previously presented techniques which shows improvement even with a small resolution level. Error bound with convergence analysis of the proposed method is also discussed in detail which shows that the error norms inversely varies with resolution level rises. Simulations describe that proposed method works well for multi-dimensional problems.

A Real-Time Large-Scale IoT Traffic Anomalies Detection System Using Auxiliary Classifier Generative Adversarial Network

Muhammad Nauman Hameed

Institute of Computing, Kohat University of Science and Technology Pakistan

Abstract:

A Real-Time IoT's Attacks Detection System for Large-Scale Anomalies Using Auxiliary Classifier Generative Adversarial Networks (AC-GAN) Muhammad Nauman Hameed, Muhammad Muneer Umar, Muhammad Irfan Uddin Institute of Computing, Kohat University of Science and Technology, Kohat 26000, Pakistan Abstract – The Internet of Things (IoT) refers to a vast network of interconnected devices that communicate and exchange data. With the exponential growth of IoT, the resulting massive traffic has created a complex digital environment. This increased connectivity introduces severe security challenges, exposing IoT networks to a variety of threats. Distinguishing between legitimate and malicious traffic is crucial for maintaining robust network security. Anomaly detection systems, including signature-based approaches, are designed to identify unusual traffic but often rely on predefined patterns, making it difficult to adapt to new and evolving threats. Although existing machine learning (ML) based systems offer greater adaptability, they are often ineffective when trained with imbalanced traffic records, which can skew detection results and reduce overall accuracy. To address these challenges, this study proposes an AI-based methodology utilizing Auxiliary Classifier Generative Adversarial Networks (AC-GAN). The model leverages AC-GAN's generative capabilities to address data imbalance and detect anomalous traffic patterns in real time. By analyzing IoT traffic datasets, such as CICIoT2023 and Bot-IoT, the approach aims to reduce false positives and negatives, significantly improving anomaly detection accuracy. This deep learning-based solution efficiently handles large-scale datasets and generates synthetic data for anomaly detection, offering a scalable and reliable approach to securing IoT networks.

Keywords: IOT SECURITY, ANOMALY DETECTION, AC-GAN, TRAFFIC PATTERNS, AI-BASED MODEL, DEEP LEARNING

Incentivizing Cooperation and Community Formation in Social Internet of IoT Networks: A study on Virtual Currencies and Social Behaviors

Saood Iqbal, Shafiullah Khan, Muhammad Muneer Umar

Institute of Computing, Kohat University of Science and Technology Pakistan

Abstract:

The Internet of Things (IoT) is an evolving paradigm that connects everyday objects, enabling them to communicate and perform collaborative tasks. Building on this concept, the Social Internet of Things (SIoT) introduces a layer of social connectivity, wherein objects establish and maintain social relationships with one another, creating a dynamic network of socially connected devices. These connections allow for the spontaneous formation of social communities among objects, influenced by social interactions and ties. However, in SIoT networks, autonomous nodes with individual objectives may act selfishly, refusing to forward data packets, which can compromise overall network performance. This paper investigates the role of virtual currencies as a mechanism to incentivize cooperation among selfish nodes, encouraging them to contribute to network functions. Additionally, it examines the process of community formation within the SIoT environment, where social behaviors and relationships shape network structures. Experimental results demonstrate that both virtual currency incentives and structured community formation significantly enhance network performance, improving connectivity and data transmission quality in SIoT networks.